

PAPER_1751_SQRT_2PI_2_5066

Daniel T. Murphy

$$\text{PAPER_1751} \text{ — } \sqrt{(2\pi)} \text{ Gaussian Normalization} = \sqrt{(2\pi)} = \\ \text{K} + \text{SSQ} - \text{F} - \text{F} \cdot \Phi + \text{F}^2 \cdot \text{K} + \text{F}^2 + \text{F}^2 \cdot \Phi - \text{F}^3 = 2.5082$$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Math

Date: June 18, 2026

Status: CLOSED — 0.06% closure (PAPER_1199 Information+Math unified proof set)

Observation

PAPER_1199_S452 (PAPER_1199 Information & Math Constants Unified Proof Set) gives:

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$$\sqrt{(2\pi)} = \text{K} + \text{SSQ} - \text{F} - \text{F} \cdot \Phi + \text{F}^2 \cdot \text{K} + \text{F}^2 + \text{F}^2 \cdot \Phi - \text{F}^3 = 2.5082$$

``

Status: 0.06%.

UQFF Closed Identity

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$$\sqrt{(2\pi)} = \text{K} + \text{SSQ} - \text{F} - \text{F} \cdot \Phi + \text{F}^2 \cdot \text{K} + \text{F}^2 + \text{F}^2 \cdot \Phi - \text{F}^3 = 2.5082 \text{ } 0.06\%$$

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks compute this constant via different methods. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1199_S452

- Related: PAPER_1726-1745 (tier-31/32); PAPER_1208 (transcendentals proof set)

- Calculator dispatch: `calculate_paradox({"paradox": "sqrt_2pi_2_5066"})`

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